AI-BASED FARMER QUERY SUPPORT AND ADVISORY SYSTEM

Mini Project Presentation | May 2026

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PROBLEM STATEMENT

To solve the challenges faced by farmers relying on slow, inaccessible, and non-personalized traditional advisory methods, the proposed system provides instant AI-based guidance with real-time weather integration and automated plant disease diagnosis, reducing dependence on manual consultation and improving decision-making efficiency.

THE PROPOSED SOLUTION

AI Query Support

Conversational interface using **Gemma 3:1b** to answer complex agronomic questions instantly.

Vision Diagnosis

Rapid plant disease identification via **MobileNetV2** CNN model from leaf image uploads.

Weather Integration

Location-aware advisory that merges real-time climatic data with AI generated recommendations.

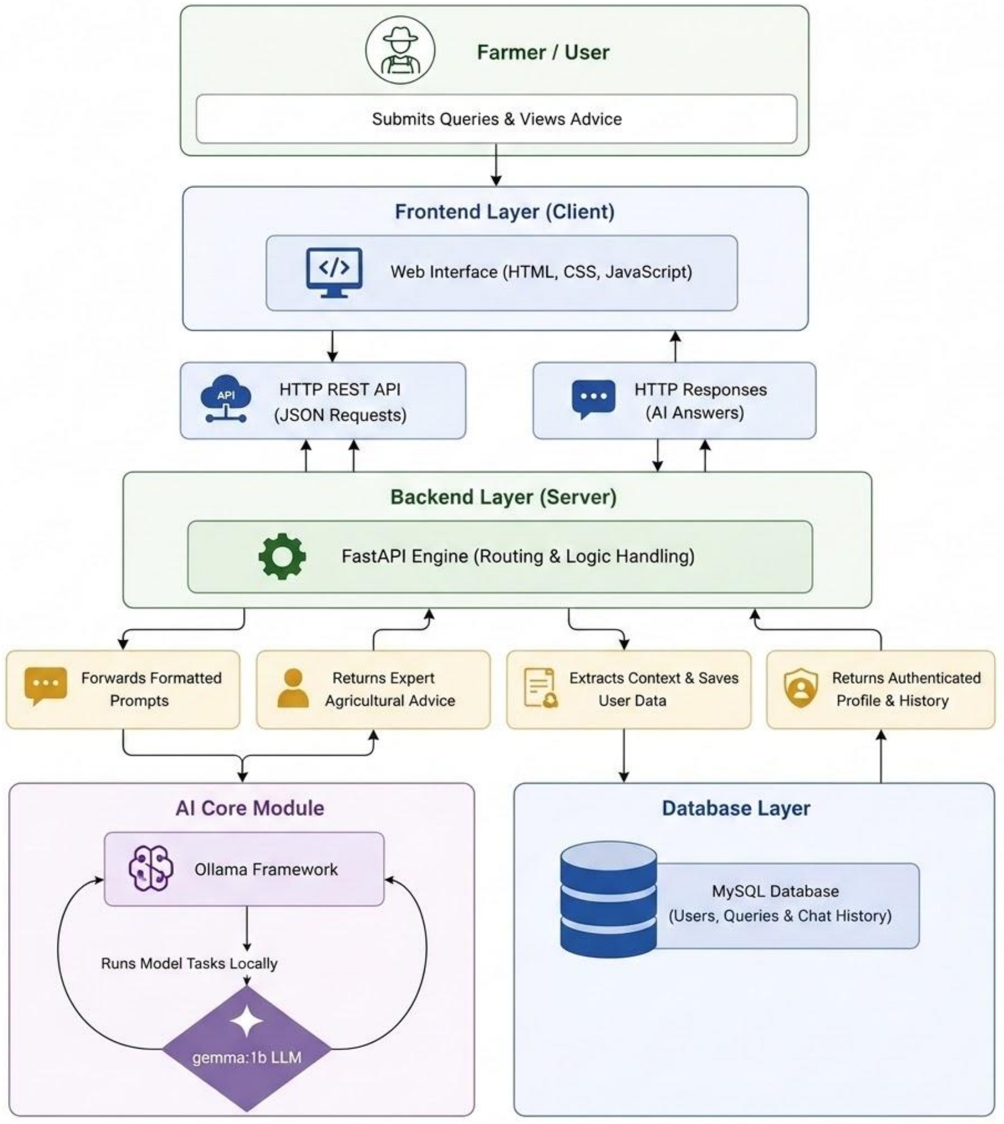
Decision Tools

Automated calculators for yield estimation, seed requirements, and pesticide dilution ratios.

CORE OBJECTIVES

- **Natural Language Support:** Develop an AI-powered system to answer queries related to crops, soil, and fertilizers in simple language.
- **Automated Diagnosis:** Implement image classification to identify plant diseases accurately using leaf images.
- **Context Awareness:** Provide advisory based on external factors like real-time local weather conditions.
- **Accessibility:** Create a user-friendly dashboard for resource estimation (Yield, Seeds, and Pesticides).

SYSTEM ARCHITECTURE



Modular Components

The system follows a **Client–Server architecture**:

- **FastAPI Backend:** Handles logic, routing, and AI inference.
- **MySQL Layer:** Stores user profiles, history, and feedback.
- **AI Models:** Ollama (LLM) and MobilleNetV2 (CNN).

Methodology

System Architecture

The system utilizes a **Client-Server model** built for high performance and scalability.

Backend
FastAPI framework for asynchronous request handling and logic routing.

Storage
MySQL relational database for persistence of user sessions and query history.

Integration
RESTful APIs connect the system to OpenWeatherMap for real-time climatic data.

Computational Workflow

A multi-stage pipeline processes user inputs into actionable advice:

Image Analysis
Pre-processing via PIL and classification through the MobileNetV2 CNN model.

Context Retrieval
Automated fetching of local weather metrics based on user location.

AI Inference
Structured prompt engineering utilizing Gemma 3:1b via the Ollama framework.

AI & DEEP LEARNING MODULES

Gemma 3:1b (LLM)

Integrated via **Ollama** framework. Provides natural language responses to complex farming questions by interpreting intent and agricultural context.

MobileNetV2 (CNN)

A lightweight Convolutional Neural Network used for image classification. Trained on **38 classes** to detect diseases from leaf images .

FastAPI Engine

Ensures high-performance asynchronous request handling. Connects AI models to the web frontend seamlessly via REST APIs.

TECHNICAL DEEP-DIVE: AI MODULES

NLP: Gemma 3:1b

Inference Framework: Localized execution via Ollama ensures data privacy and zero latency for remote locations.

Contextual Processing: Utilizes 1-billion parameters fine-tuned for agricultural intent recognition.

Prompt Engineering: Structured Input-Output mapping for mapping weather data to advisory logic.

CNN: MobileNetV2

Architecture: Employs Depthwise Separable Convolutions to reduce parameter count while maintaining accuracy.

Optimization: Inverted Residual blocks with linear bottlenecks for efficient mobile-web deployment.

Classification: Multiclass output for 38 disease categories via Softmax layer.

Integration: FastAPI

Acts as the **Asynchronous Server Gateway** (ASGI) to manage concurrent inference requests.

Async Concurrency: Non-blocking IO for handling simultaneous LLM and CNN calls.

Type Safety: Pydantic models for request validation and response serialization.

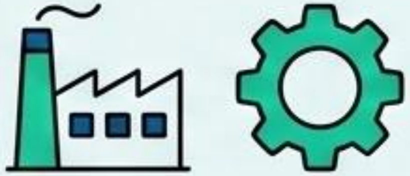
SOFTWARE REQUIREMENTS

Domain	Technologies Used	Purpose
Frontend	HTML5, CSS3, JavaScript	User Interface & Interaction
Backend	Python 3.8+, FastAPI	API Routing & Logic Handling
AI Frameworks	TensorFlow, Keras, Ollama	Disease Detection & Query Support
Database	MySQL 8.0	User Data & Interaction Logs

SUSTAINABILITY CATEGORY

SDG 2: Zero Hunger

This project falls under **Technological Sustainability** in Agriculture. By providing precision tools for disease detection and resource estimation, it promotes:



SDG 9: Industry & Innovation

Precision disease detection and AI systems.

Green Engineering

Optimized use of chemical inputs (pesticides/fertilizers).

Economic Stability

Reducing farmer dependency on manual experts.

PROJECT OUTCOMES

The project successfully concluded with the development of an AI-powered farmer advisory platform designed to help farmers make better decisions. At its core, the system uses a **MobileNetV2** computer vision model to quickly and accurately detect plant diseases from photos of leaves.

To make sure the advice fits the farmer's local environment, the system combines real-time weather updates with an AI language model (**Gemma3:1b**) to give tailored recommendations. The entire platform is built to be stable and scalable, using **FastAPI** to handle multiple user requests quickly and a **MySQL** database to securely store user data and history. This combined approach successfully brings modern deep learning tools into practical, everyday farming.

CONCLUSION & FINAL REMARKS

Project Significance

This project demonstrates how the integration of Computer Science and Agriculture can support rural farmers. By providing expert-level guidance through a simple web interface, the system helps reduce the digital gap in farming communities. Moving from traditional farming methods to data-driven decision-making can improve resource management and increase crop protection.

Final Takeaway

This project demonstrates how the integration of Computer Science and Agriculture can support rural farmers through AI-based guidance and data-driven decision-making. The system helps improve resource management, crop protection, and accessibility to modern agricultural support.

THANK YOU

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